

Full Molecular Hamiltonian

From Classical to Quantum: the exact non-relativistic quantization

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Roadmap: four stages

We build the **exact** non-relativistic molecular Hamiltonian in four stages:

1. **Classical** — lab-frame H ; one point transformation separates the centre of mass *and* produces the mass polarization.
2. **Body-Fixed** — co-rotating frame + Routhian $\Rightarrow$ classical molecular H_{int} .
3. **Quantize** — Podolsky rule from **coordinate derivatives alone**: metric $\rightarrow$ measure $\rightarrow$ inverse $\rightarrow$ Watson pseudopotential. No angular-momentum algebra.
4. **Quantum** $\hat{H}$ — the exact non-relativistic molecular Hamiltonian.

Outline

Motivation

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Mass-centered coordinates

Body-fixed frame and the Routhian

The quantization problem

The ro-vibrational geometry

Coordinate-native quantization

The complete quantum Hamiltonian

Summary

Motivation

Why is the molecular Hamiltonian non-trivial?

The lab Hamiltonian couples *all* particles through the Coulomb potential:

$$H = \sum_j \frac{\mathbf{P}_j^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V.$$

To do molecular physics we want to separate

$$\underbrace{\text{translation}}_{\text{COM}} \quad | \quad \underbrace{\text{rotation}}_{\text{Euler angles } \Omega} \quad | \quad \underbrace{\text{vibration}}_{\text{normal coords } Q} \quad | \quad \underbrace{\text{electrons}}_{\vec{r}_i}.$$

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The catch

Rotation + vibration are *curvilinear* coordinates. Naively quantizing $p \rightarrow -i\hbar \partial$ is **not** Hermitian there. Getting it right forces the Podolsky construction.

What we will not approximate

We keep everything exact until the very end:

- **No** Born–Oppenheimer / clamped nuclei.
- **No** neglect of *mass polarization*.
- **No** harmonic assumption.

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The *only* ingredients are:

1. non-relativistic dynamics;
2. the Eckart frame (a *convention*, not an approximation);
3. parametrizing the nuclear shape by $3N_{\text{nuc}} - 6$ normal coordinates (exact).

The endpoint is the exact quantum $\hat{H}_{\text{int}}$ — the rigorous starting point

Classical Cartesian Hamiltonian

The Lagrangian in the lab frame

N_{nuc} nuclei ($M_j, Z_j e, \mathbf{R}_j$) and N_{elec} electrons ($m_e, -e, \mathbf{r}_i$), all in the space-fixed frame; SI units.
The Lagrangian is kinetic minus potential:

$$L = \sum_j \frac{1}{2} M_j \dot{\mathbf{R}}_j^2 + \sum_i \frac{1}{2} m_e \dot{\mathbf{r}}_i^2 - V(\mathbf{R}_j, \mathbf{r}_i).$$

Differentiate L with respect to each velocity to get the **canonical momenta**:

$$\mathbf{P}_j \equiv \frac{\partial L}{\partial \dot{\mathbf{R}}_j} = M_j \dot{\mathbf{R}}_j, \quad \mathbf{p}_i \equiv \frac{\partial L}{\partial \dot{\mathbf{r}}_i} = m_e \dot{\mathbf{r}}_i.$$

Canonical brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk} \delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij} \delta_{\alpha\beta}$.

- Kinetic energy is diagonal and quadratic in Cartesian velocities: *canonical momentum* = *mass* $\times$ *velocity*, no surprises.
- This is the *only* stage of the whole paper where canonical and mechanical momentum coincide.

Legendre transform to the Hamiltonian

The three-move recipe (we reuse it, less trivially, in a moment):

1. **Define** $p_q \equiv \partial L / \partial \dot{q}$ for every coordinate.
2. **Invert** for every velocity as a function of the momenta.
3. **Assemble** $H \equiv \sum_q p_q \dot{q} - L$ and substitute.

Here step 2 is immediate, $\dot{\mathbf{R}}_j = \mathbf{P}_j / M_j$, $\dot{\mathbf{r}}_i = \mathbf{p}_i / m_e$; step 3 gives

$$H = \underbrace{\sum_j \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_i \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \quad (1.1)$$

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{i \neq j} \frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j} \frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{r_{ij}} \right). \quad (1.2)$$

Everything that follows is an exact rewriting of (1.1)–(1.2).

Mass-centered coordinates

Two centers of mass, two roles

The translational coordinate and the internal reference point **need not be the same**. Define

$$M_N = \sum_j M_j, \quad M_{\text{tot}} = M_N + N_{\text{elec}} m_e,$$

$$\mathbf{R}_{\text{ncm}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j, \quad \mathbf{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \left(\sum_j M_j \mathbf{R}_j + m_e \sum_i \mathbf{r}_i \right), \quad (\text{A.2, A.3})$$

and refer every internal coordinate — nuclear and electronic — to the *nuclear* CoM:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{ncm}}, \quad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{ncm}}, \quad \sum_j M_j \mathbf{R}'_j = 0. \quad (\text{A.4})$$

The mixed choice

Translation = total CoM $\Rightarrow$ correct total mass, and (we will see) the COM–internal cross terms cancel identically.

Internals referred to the nuclear CoM $\Rightarrow$ Born–Oppenheimer-friendly coordinates, V form-invariant, and the mass polarization appears *by itself*.

Inverting: every particle rides a shared drift

Solve (A.3) for the nuclear CoM, then invert (A.4) for the lab positions:

$$\mathbf{R}_{\text{ncm}} = \mathbf{R}_{\text{cm}} - \frac{m_e}{M_{\text{tot}}} \sum_l \mathbf{r}'_l, \quad (\text{A.5})$$

$$\mathbf{R}_j = \mathbf{R}_{\text{cm}} + \mathbf{R}'_j - \frac{m_e}{M_{\text{tot}}} \sum_l \mathbf{r}'_l, \quad \mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}'_i - \frac{m_e}{M_{\text{tot}}} \sum_l \mathbf{r}'_l. \quad (\text{A.6})$$

Both families ride on the *same* drift — the nuclear-CoM velocity:

$$\boxed{\mathbf{D} \equiv \dot{\mathbf{R}}_{\text{ncm}} = \dot{\mathbf{R}}_{\text{cm}} - \frac{m_e}{M_{\text{tot}}} \sum_l \dot{\mathbf{r}}'_l, \quad \dot{\mathbf{R}}_j = \mathbf{D} + \dot{\mathbf{R}}'_j, \quad \dot{\mathbf{r}}_i = \mathbf{D} + \dot{\mathbf{r}}'_i.} \quad (\text{A.7})$$

Every lab velocity = internal velocity + one shared drift $\mathbf{D}$ — passengers on a train.

Kinetic energy in velocity form: the mass polarization appears

Substitute (A.7) into $T = \sum_j \frac{1}{2} M_j \dot{\mathbf{R}}_j^2 + \sum_i \frac{1}{2} m_e \dot{\mathbf{r}}_i^2$. The nuclear cross term dies by $\sum_j M_j \dot{\mathbf{R}}_j' = 0$; the electronic one does *not*:

$$T_N = \frac{1}{2} M_N \mathbf{D}^2 + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2, \quad T_e = \frac{1}{2} N_{\text{elec}} m_e \mathbf{D}^2 + m_e \mathbf{D} \cdot \sum_i \dot{\mathbf{r}}_i' + \frac{1}{2} m_e \sum_i \dot{\mathbf{r}}_i'^2. \quad (\text{A.8, A.9})$$

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Expand $\mathbf{D}$ in the two $\mathbf{D}$ -dependent pieces (A.11a,b): the $\dot{\mathbf{R}}_{\text{cm}} \cdot \sum \dot{\mathbf{r}}'$ cross terms cancel **identically**, and the quadratic pieces combine to one *negative* remainder:

$$T = \frac{1}{2} M_{\text{tot}} \dot{\mathbf{R}}_{\text{cm}}^2 + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2 + \frac{1}{2} m_e \sum_i \dot{\mathbf{r}}_i'^2 - \frac{m_e^2}{2M_{\text{tot}}} \left(\sum_i \dot{\mathbf{r}}_i' \right)^2. \quad (\text{A.12})$$

The CoM decouples with the *correct* mass M_{tot} ; the electrons carry a negative collective correction — the mass polarization in disguise.

Canonical momenta: $\partial L / \partial \dot{q}$, and the tricky one

With $L = T - V$ (V velocity-independent), differentiate the boxed T :

$$\mathbf{P}_{\text{cm}} = \frac{\partial L}{\partial \dot{\mathbf{R}}_{\text{cm}}} = M_{\text{tot}} \dot{\mathbf{R}}_{\text{cm}}, \quad \mathbf{P}'_j = \frac{\partial L}{\partial \dot{\mathbf{R}}'_j} = M_j \dot{\mathbf{R}}'_j. \quad (\text{A.13, A.17})$$

The electrons take care: *two* terms of T involve $\dot{\mathbf{r}}'_i$ (its own kinetic term, *and* the mass-polarization sum), so

$$\mathbf{p}'_i \equiv \frac{\partial L}{\partial \dot{\mathbf{r}}'_i} = m_e \dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\text{tot}}} \sum_l \dot{\mathbf{r}}'_l \neq m_e \dot{\mathbf{r}}'_i. \quad (\text{A.14})$$

Canonical $\neq$ mechanical: moving *one* electron drags the whole molecule's CoM, and $\mathbf{r}'_i$ is measured from the (recoiling) nuclear CoM — every electron's momentum carries a shared correction.

Inverting the electron relation

Sum (A.14) over all electrons; with $M_{\text{tot}} - N_{\text{elec}} m_e = M_N$,

$$\sum_i \mathbf{p}'_i = m_e \left(1 - \frac{N_{\text{elec}} m_e}{M_{\text{tot}}} \right) \sum_l \dot{\mathbf{r}}'_l = \frac{m_e M_N}{M_{\text{tot}}} \sum_l \dot{\mathbf{r}}'_l. \quad (\text{A.15})$$

Solve for $\dot{\mathbf{r}}'_i$:

$$\boxed{\dot{\mathbf{r}}'_i = \frac{\mathbf{p}'_i}{m_e} + \frac{1}{M_N} \sum_l \mathbf{p}'_l.} \quad (\text{A.16})$$

Each electron's velocity = its own momentum/ m_e + one shared recoil — the momentum-space mirror of the shared drift $\mathbf{D}$.

Legendre transform: kinetic energy in momentum form

CoM and nuclear terms are immediate: $\frac{1}{2} M_{\text{tot}} \dot{\mathbf{R}}_{\text{cm}}^2 = \mathbf{P}_{\text{cm}}^2 / 2M_{\text{tot}}$, $\frac{1}{2} M_j \dot{\mathbf{R}}_j'^2 = \mathbf{P}_j'^2 / 2M_j$.
 Substituting (A.16) into the electron sector (A.19, A.20) and collecting the $(\sum_i \mathbf{p}_i')^2$ coefficients,

$$\frac{1}{M_N} + \frac{N_{\text{elec}} m_e - M_{\text{tot}}}{2M_N^2} = \frac{1}{M_N} - \frac{M_N}{2M_N^2} = +\frac{1}{2M_N}. \quad (\text{A.21})$$

The negative velocity-form correction has become the positive mass polarization:

$$T = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (\text{A.22})$$

The Hamiltonian — and why the sign flips

Both internal families share the reference point $\mathbf{R}_{\text{ncm}}$, so $V(\mathbf{R}'_j, \mathbf{r}'_i)$ is unchanged in form. Add it to (A.22):

$$H = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \underbrace{\sum_j \frac{\mathbf{P}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}'_i{}^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \mathbf{p}'_i \right)^2}_{H_{\text{int}} \text{ (2.13)}} + V(\mathbf{R}'_j, \mathbf{r}'_i). \quad (2.12 \equiv \text{A.23})$$

Why the sign flips

The electron block of T is the quadratic form $\frac{1}{2} \dot{\mathbf{r}}'^T \mathbf{A} \dot{\mathbf{r}}'$ with $\mathbf{A} = m_e (\mathbb{1} - \frac{m_e}{M_{\text{tot}}} \mathbf{J})$ ($\mathbf{J}$ all-ones); the Legendre transform replaces $\mathbf{A} \rightarrow \mathbf{A}^{-1} = \frac{1}{m_e} (\mathbb{1} + \frac{m_e}{M_N} \mathbf{J})$ (A.25, A.26) — same physics, opposite side of the transform.

Exact for any $\mathbf{P}_{\text{cm}}$: no rest frame, no boost, no cross-coupling — agrees with the paper's generating-function route (Sec. 2). H_{int} is what we carry into the body-fixed frame.

Body-fixed frame and the Routhian

Body-fixed frame: rotate with the nuclei

Introduce a rotation $\mathbf{S}(\Omega)$ (Euler angles $\Omega = \phi, \theta, \chi$) and body-fixed (BF) coordinates:

$$\mathbf{R}'_j = \mathbf{S}(\Omega) \bar{\mathbf{R}}_j(Q), \quad \mathbf{r}'_i = \mathbf{S}(\Omega) \bar{\mathbf{r}}_i. \quad (3.1)$$

The *Eckart* conditions fix $\mathbf{S}$ from the *nuclei*; the nuclear shape is rectilinear mass-weighted **normal coordinates** Q_b :

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\text{eq}} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb} Q_b, \quad \sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}. \quad (3.1a)$$

The BF angular velocity ω is defined by

$$(\mathbf{S}^T \dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma} \omega_\gamma. \quad (3.2)$$

A Routhian that keeps electrons in momentum form

Start from H (2.12). We need an action principle for the body-fixed *nuclear* coordinates — but the electron kinetic energy + mass polarization $H_e = \sum_i \mathbf{p}'_i{}^2/2m_e + \frac{1}{2M_N}(\sum_i \mathbf{p}'_i)^2$ is a *rotational scalar*. So we Legendre-transform only the COM and nuclear momenta and **keep the electrons in momentum form**, giving the **Routhian**

$$\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}'_j|^2 - H_e(\mathbf{p}'_i) - V. \quad (3.6)$$

Why a Routhian?

$\mathcal{R}$ is a Lagrangian in the nuclear/COM coordinates $(\mathbf{R}_{\text{cm}}, \mathbf{R}'_j)$ and *minus* a Hamiltonian in the electrons $(\mathbf{r}'_i, \mathbf{p}'_i)$: the electrons ride along as the scalar H_e , never converted to velocity form and back.

Body-fixed Routhian

H_e is built from $\mathbf{p}'_j \cdot \mathbf{p}'_j$; under the passive rotation $\bar{\mathbf{p}}_i \equiv \mathbf{S}^T \mathbf{p}'_i$ it is *unchanged* in form. Substituting the BF velocities $\dot{\mathbf{R}}'_j = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j)$,

$$\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 - \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} - \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 - V. \quad (3.7)$$

Only the *nuclear* kinetic term carries $\boldsymbol{\omega}$; V is now manifestly $\mathbf{S}$ -independent.

Expand the nuclear term

The vector identity (for any $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega}$)

$$\frac{m}{2} |\boldsymbol{\omega} \times \mathbf{a} + \dot{\mathbf{a}}|^2 = \underbrace{\frac{m}{2} \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}})}_{\text{Coriolis}} + \underbrace{\frac{m}{2} |\dot{\mathbf{a}}|^2}_{\text{vibrational}} \quad (3.9c)$$

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applied to the nuclei ($m \rightarrow M_j$, $\mathbf{a} \rightarrow \bar{\mathbf{R}}_j$) gives the assembled Routhian

$$\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \underbrace{\frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N}_{\text{nuclear}} - H_e - V. \quad (3.13)$$

The rotational kernel is the *nuclear* inertia $\mathbf{I}_N$ alone — no electronic inertia, because the electrons were never expanded in $\boldsymbol{\omega}$.

Canonical momenta and the total angular momentum

Nuclear angular momentum (from $\partial\mathcal{R}/\partial\omega$):

$$\mathbf{J}_N = \mathbf{I}_N\omega + \mathbf{L}_N^{\text{vib}}. \quad (3.15)$$

The BF frame rotates nuclei *and* electrons, so the **total** angular momentum is the sum [$\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$]:

$$\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}} = \mathbf{I}_N\omega + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_{\text{elec}}. \quad (3.16)$$

Vibrational momentum, with the **Coriolis vectors** σ_b :

$$P_b = \omega \cdot \sigma_b + \dot{Q}_b, \quad \sigma_b = \sum_a \zeta_{ab} Q_a, \quad \zeta_{ab} = \sum_j \mathbf{I}_{ja} \times \mathbf{I}_{jb}. \quad (3.17\text{a,b})$$

The key identity

Subtract the electronic part from (3.16):

$$\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N(Q) \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}. \quad (3.19)$$

Why this matters

The right side is *purely nuclear*: the rotational kernel is $\mathbf{I}_N$, **not** the full inertia. All electronic + mass-polarization content sits on the left, in the kinematic shift $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}}$.

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Cross-check: the velocity-form route instead generates the electronic inertia $\mathbf{I}_e, \mathbf{I}_X$ in *both* $\mathbf{J}$ and $\mathbf{L}_{\text{elec}}$ — they cancel exactly, reproducing (3.19). The Routhian makes the cancellation automatic.

Watson reciprocal inertia

Invert (3.19) for ω : write $\mathbf{L}_N^{\text{vib}} = \boldsymbol{\pi} - \boldsymbol{\sigma}\boldsymbol{\sigma}^T\omega$ with

$$\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b \quad (\text{field-free vibrational ang. mom.}), \quad \boldsymbol{\sigma}\boldsymbol{\sigma}^T = \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T. \quad (3.20a)$$

Then $\mathbf{J} - \mathbf{L}_{\text{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)\omega + \boldsymbol{\pi}$, so

$$\boxed{\omega = \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}), \quad \boldsymbol{\mu} \equiv (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)^{-1}.} \quad (3.20)$$

$\boldsymbol{\mu}$ is the **Watson reciprocal-inertia tensor** — built from *nuclear* quantities only, with no Wilson G -matrix.

The exact classical molecular Hamiltonian

Legendre back; the rovibrational kinetic energy becomes

$$\frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2:$$

$$\begin{aligned} H_{\text{int}} = & \frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(Q)(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2 \\ & + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 + V_{NN} + V_{Ne} + V_{ee}. \end{aligned} \quad (3.22)$$

Exact, classical, in BF + Eckart coordinates. **Now we quantize.**

The quantization problem

Why naive quantization fails in curvilinear coordinates

Label the curvilinear coordinates and write the kinetic energy as a quadratic form:

$$\{q^A\} = \{\phi, \theta, \chi, Q_1, \dots, Q_f\}, \quad f = 3N_{\text{nuc}} - 6, \quad T = \frac{1}{2} g^{AB}(q) p_A p_B. \quad (4.1, 4.3)$$

Three objects control everything: the metric g_{AB} , its determinant g , and its inverse g^{AB} .

The problem, in one dimension

On $L^2(w(q) dq)$ with a non-constant weight, integration by parts gives

$$(-i\hbar \partial_q)^\dagger = -i\hbar(\partial_q + w'/w) \neq -i\hbar \partial_q. \quad (4.4)$$

The curved measure $\sqrt{g} d^n q$ plays the role of w : naive $p_A \rightarrow -i\hbar \partial_A$ is **non-Hermitian**.

The Podolsky / Laplace–Beltrami prescription

The kinetic operator Hermitian with respect to $\sqrt{g} d^n q$ by construction:

$$\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{g}} \partial_A (\sqrt{g} g^{AB} \partial_B), \quad g = \det g_{AB}. \quad (4.5)$$

One integration by parts gives the manifestly symmetric (and positive) form

$$\int \varphi^* \hat{T} \psi \sqrt{g} d^n q = \frac{\hbar^2}{2} \int (\partial_A \varphi)^* g^{AB} (\partial_B \psi) \sqrt{g} d^n q. \quad (4.6)$$

For $g_{AB} = \delta_{AB}$ it reduces to $-\frac{\hbar^2}{2} \nabla^2$.

The whole program of this stage

Compute g_{AB} , $\det g$, g^{AB} for the coordinates (4.1) — *explicitly* — then expand (4.5) block by block. Hermiticity is automatic at every step.

The ro-vibrational geometry

Euler-angle kinematics, explicitly

zyz convention $\mathbf{S} = R_z(\phi)R_y(\theta)R_z(\chi)$ (4.7). Differentiating $\mathbf{S}$ gives $\boldsymbol{\omega} = \mathbf{E}(\boldsymbol{\Omega})\dot{\boldsymbol{\Omega}}$ and, solving the linear system, $\dot{\boldsymbol{\Omega}} = \mathbf{E}^{-1}\boldsymbol{\omega}$:

$$\mathbf{E} = \begin{pmatrix} -\sin\theta\cos\chi & \sin\chi & 0 \\ \sin\theta\sin\chi & \cos\chi & 0 \\ \cos\theta & 0 & 1 \end{pmatrix}, \quad \mathbf{E}^{-1} = \begin{pmatrix} -\frac{\cos\chi}{\sin\theta} & \frac{\sin\chi}{\sin\theta} & 0 \\ \sin\chi & \cos\chi & 0 \\ \frac{\cos\theta\cos\chi}{\sin\theta} & -\frac{\cos\theta\sin\chi}{\sin\theta} & 1 \end{pmatrix} \quad (4.10, 4.12)$$

with

$$\det \mathbf{E} = -\sin\theta. \quad (4.11)$$

The $1/\sin\theta$ entries are the Euler coordinate singularity at the poles; they are compensated by $|\det \mathbf{E}| = \sin\theta$ in the volume element. (4.10)–(4.12) are the only Euler-angle facts used from here on.

The ro-vibrational metric tensor

The classical BF kinetic energy $2T_{\text{rv}} = \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + 2\boldsymbol{\omega}^T \sum_b \boldsymbol{\sigma}_b \dot{Q}_b + \sum_b \dot{Q}_b^2$ (4.13), re-expressed in the *coordinate* velocities via $\boldsymbol{\omega} = \mathbf{E}\dot{\boldsymbol{\Omega}}$, defines the metric:

$$g_{AB} = \begin{pmatrix} \mathbf{E}^T \mathbf{I}_N \mathbf{E} & \mathbf{E}^T \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T \mathbf{E} & \mathbb{1}_f \end{pmatrix} \quad (4.15)$$

- Euler–Euler block: inertia pulled back through the kinematic matrix.
- Euler–mode block: Coriolis coupling $\boldsymbol{\sigma}_b = \sum_a \zeta_{ab} Q_a$ (constant ζ 's).
- mode–mode block: *identity* — the gift of mass-weighted rectilinear normal coordinates.

Every entry is an explicit function of (θ, χ, Q) .

Electrons: flat, constant, decoupled from the measure

Including electron velocities, the full mass matrix has a *constant* electron block D. Eliminating it (Schur complement = completing the square) removes all ω -dependence:

$$CD^{-1}C^T = \mathbf{I}_e - \mathbf{I}_X \implies \det M = \underbrace{\det D}_{\text{const}} \cdot \det \begin{pmatrix} \mathbf{I}_N & \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T & \mathbb{1}_f \end{pmatrix} \quad (4.17)$$

— the same $\mathbf{I}_e/\mathbf{I}_X$ cancellation as the classical cross-check of § 3.6.

- Electron measure is the flat $\prod_i d^3\bar{\mathbf{r}}_i$; $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is Hermitian on it.
- $\hat{L}_{\text{elec},\alpha}$ is Hermitian *without symmetrization*: its vector field is divergence-free, $\sum_{i\gamma} \partial_{\bar{r}_{i\gamma}} (\epsilon_{\alpha\beta\gamma} \bar{r}_{i\beta}) = 0$ (4.18).

The entire quantization problem lives in the ro-vibrational sector; electrons enter only via the flat kinetic operator and the shift $\hat{J}_\alpha \rightarrow \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha}$.

Determinant and volume element

Block row reduction gives $\det \begin{pmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{b}^T & \mathbb{1} \end{pmatrix} = \det(\mathbf{a} - \mathbf{b}\mathbf{b}^T)$ (4.19). Applied to (4.15):

$$\det g_{AB} = \sin^2 \theta \gamma(Q), \quad \gamma \equiv \det \mathbf{I}', \quad \mathbf{I}' \equiv \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T. \quad (4.20)$$

The Watson-modified inertia appears by itself

The determinant is governed by $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T$ — *not* the bare $\mathbf{I}_N$: the Coriolis coupling reduces the rotational determinant. It is the same $\mathbf{I}'$ whose inverse $\boldsymbol{\mu}$ is the classical rotational kernel (3.20).

$$\sqrt{g} d^{3+f} q = \underbrace{\sin \theta d\phi d\theta d\chi}_{\text{Haar measure on } SO(3)} \cdot \underbrace{\sqrt{\gamma(Q)} \prod_b dQ_b}_{\text{rot.-vib. Jacobian}} \cdot \underbrace{\prod_i d^3 \mathbf{r}_i}_{\text{flat}} \quad (4.21)$$

The inverse metric

Claim, then *verify* by block multiplication using only $\mu \mathbf{l}' = \mathbb{1}$ [(4.23)]:

$$g^{AB} = \begin{pmatrix} E^{-1} \mu E^{-T} & -E^{-1} \mu \sigma \\ -\sigma^T \mu E^{-T} & \mathbb{1}_f + \sigma^T \mu \sigma \end{pmatrix}, \quad \mu = (\mathbf{l}_N - \sigma \sigma^T)^{-1}. \quad (4.22)$$

Two inversions, one computation

The Watson reciprocal inertia μ — introduced classically by inverting the momentum–velocity relation (3.20) — reappears as the geometric inverse of the metric.

e.g. top-left: $E^T \mathbf{l}_N E \cdot E^{-1} \mu E^{-T} - E^T \sigma \cdot \sigma^T \mu E^{-T} = E^T (\mathbf{l}_N - \sigma \sigma^T) \mu E^{-T} = \mathbb{1}_3$. $\checkmark$

Coordinate-native quantization

$\hat{J}_\alpha$: defined, not postulated

The expansion of (4.5) keeps producing $(E^{-1})_{s\alpha} \partial_{\Omega_s}$. Name it:

$$\hat{J}_\alpha \equiv -i\hbar \sum_s (E^{-1})_{s\alpha} \partial_{\Omega_s} : \quad \begin{aligned} \hat{J}_1 &= -i\hbar \left[-\frac{\cos \chi}{\sin \theta} \partial_\phi + \sin \chi \partial_\theta + \frac{\cos \theta \cos \chi}{\sin \theta} \partial_\chi \right], \\ \hat{J}_2 &= -i\hbar \left[\frac{\sin \chi}{\sin \theta} \partial_\phi + \cos \chi \partial_\theta - \frac{\cos \theta \sin \chi}{\sin \theta} \partial_\chi \right], \\ \hat{J}_3 &= -i\hbar \partial_\chi. \end{aligned} \quad (4.24)$$

The logic

(4.24) is a *definition* — shorthand for three explicit combinations of Euler-angle derivatives. We will **never** compute a commutator $[\hat{J}_\alpha, \hat{J}_\beta]$ or use any angular-momentum algebra.

Classical correspondence: $\frac{1}{2} p_A g^{AB} p_B = \frac{1}{2} \mathbf{J}^T \boldsymbol{\mu} \mathbf{J} - \mathbf{J}^T \boldsymbol{\mu} \boldsymbol{\pi} + \frac{1}{2} \boldsymbol{\pi}^T \boldsymbol{\mu} \boldsymbol{\pi} + \frac{1}{2} \sum_b P_b^2$ — exactly the nuclear kernel of (3.22). (4.25)

The divergence identity: the workhorse of this section

The weighted divergence of each column of E^{-1} vanishes — computed term by term from (4.12):

$$\begin{aligned}\alpha = 1 : \quad & \partial_\phi(-\cos \chi) + \partial_\theta(\sin \theta \sin \chi) + \partial_\chi(\cos \theta \cos \chi) = 0 + \cos \theta \sin \chi - \cos \theta \sin \chi = 0, \\ \alpha = 2 : \quad & \partial_\phi(\sin \chi) + \partial_\theta(\sin \theta \cos \chi) + \partial_\chi(-\cos \theta \sin \chi) = 0 + \cos \theta \cos \chi - \cos \theta \cos \chi = 0, \\ \alpha = 3 : \quad & \partial_\chi(\sin \theta) = 0.\end{aligned}\tag{4.26}$$

Hence each $\hat{J}_\alpha$ is **Hermitian on the Haar measure** $\sin \theta \, d\Omega$ by one integration by parts — *no symmetrization needed* (4.27).

This is the replacement

(4.26) is the coordinate-level fact that replaces every appeal to angular-momentum algebra: the vector fields generating body-frame rotations are divergence-free with respect to the Haar measure.

The Podolsky operator, block by block

Insert (4.21)–(4.22) into (4.5) and split into rotational, Coriolis, vibrational blocks (4.28):

- **Rotational:** $\mu, \sqrt{\gamma}$ pass through ∂_Ω ; the divergence identity (4.26) kills the extra term:

$$\hat{T}_{\text{rot}} = \frac{1}{2} \sum_{\alpha\beta} \mu_{\alpha\beta}(Q) \hat{J}_\alpha \hat{J}_\beta \quad \text{— Hermitian as written, no extra potential.} \quad (4.29)$$

(“A rigid rotor has no pseudopotential” — verified by three derivative computations, not by an algebra.)

- **Coriolis:** the two groups are adjoints of each other on the measure (4.21) $\Rightarrow$ Hermitian sum, automatically (4.30).
- **Vibrational:** $g^{QQ} = \mathbb{1} + \sigma^T \mu \sigma$; the curved factor $\sqrt{\gamma}$ makes the bare $\hat{P}_b = -i\hbar \partial_{Q_b}$ non-Hermitian — first-order remainders $\propto \partial_{Q_b} \ln \gamma$ appear (4.31). Removing them is the job of the rescaling.

Rescaling to the flat measure, step by step

Absorb the shape factor into the wavefunction: $\psi_W \equiv \gamma^{1/4}\psi$, so $|\psi|^2\sqrt{g}dq = |\psi_W|^2d\mu_W$
 (4.32). Conjugate the vibrational block by $\gamma^{1/4}$ (one product rule per line):

$$\begin{aligned}\partial_{Q_b}(\gamma^{-1/4}\psi_W) &= \gamma^{-1/4}[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W], \\ \gamma^{1/2}\partial_{Q_b}(\gamma^{-1/4}\psi_W) &= \gamma^{1/4}[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W], \\ \partial_{Q_b}(\gamma^{1/4}[\dots]) &= \gamma^{1/4}[\partial_{Q_b}^2\psi_W - \tfrac{1}{16}(\partial_{Q_b}\ln\gamma)^2\psi_W - \tfrac{1}{4}(\partial_{Q_b}^2\ln\gamma)\psi_W].\end{aligned}\quad (4.33)$$

The cross terms cancel; what survives is the bare $\frac{1}{2}\sum_b\hat{P}_b^2$ plus a *multiplicative* residue U_{vib}
 (4.34).

The Coriolis block: the induced $\pm\frac{i\hbar}{4}(\partial_{Q_b}\ln\gamma)$ terms are equal and opposite $\Rightarrow$ they cancel, leaving the symmetric $-\frac{1}{2}(\hat{J}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha\mu_{\alpha\beta}\hat{J}_\beta)$ with **no residue** (4.35).

Collapse to the Watson trace

The total residue $U_{\text{vib}} + U_{\text{Cor}}$ collapses through two exact Q -calculus identities. **Jacobi:**

$$\partial_{Q_b} \ln \gamma = \text{tr}(\boldsymbol{\mu} \partial_{Q_b} \mathbf{I}'), \quad \partial_{Q_b} \boldsymbol{\mu} = -\boldsymbol{\mu}(\partial_{Q_b} \mathbf{I}')\boldsymbol{\mu}. \quad (4.36)$$

Watson's closed form (4.37): $\mathbf{I}' = A(\mathbf{I}^e)^{-1}A$ with $A = \mathbf{I}^e + \frac{1}{2} \sum_b a^b Q_b$ and *constant* inertia derivatives a^b , so every derivative of $\gamma, \boldsymbol{\mu}$ reduces to constants. The gradient terms cancel; the Eckart sum rules fix the remainder:

$$\hat{U}(Q) = -\frac{\hbar^2}{8} \sum_{\alpha=1}^3 \mu_{\alpha\alpha}(Q). \quad (4.38)$$

- A pure *c*-number from the *measure* — kernel ordering never generates one.
- The electrons do not modify it ($\gamma, \boldsymbol{\mu}$ depend on Q alone; electronic Jacobian flat).
- The collapse is special to *rectilinear* normal coordinates (the closed form (4.37)).

The complete quantum Hamiltonian

Electronic kinetic operator

In BF Cartesian electron coordinates $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\mathbf{r}_i}$ is already Hermitian on the flat measure, so the electron + mass-polarization piece quantizes directly:

$$\hat{T}_e = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2\mu_e} + \frac{1}{M_N} \sum_{i<l} \hat{\mathbf{p}}_i \cdot \hat{\mathbf{p}}_l. \quad (4.39, 4.40)$$

The diagonal part renormalizes the electron mass, $\frac{1}{m_e} + \frac{1}{M_N} = \frac{1}{\mu_e}$; the off-diagonal two-electron term is the mass polarization *sensu stricto* (nuclear recoil).

The exact non-relativistic molecular Hamiltonian

Collecting the flat-measure blocks — rotational (4.29), Coriolis (4.35), vibrational, pseudopotential (4.38), electronic (4.39), and the shift $\hat{J}_\alpha \rightarrow \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha}$:

$$\begin{aligned}
 \hat{H}_{\text{int}} = & \frac{1}{2} \sum_{\alpha\beta} (\hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha) \mu_{\alpha\beta}(Q) (\hat{J}_\beta - \hat{L}_{\text{elec},\beta} - \hat{\pi}_\beta) && \text{rotation-vibration-electron} \\
 & + \frac{1}{2} \sum_b \hat{P}_b^2 && \text{vibration} \\
 & + \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 && \text{electrons + mass pol.} \\
 & - \frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}(Q) && \text{Watson pseudopotential} \\
 & + V_{NN} + V_{Ne} + V_{ee} && \text{Coulomb}
 \end{aligned} \tag{4.41}$$

on $L^2(d\mu_W)$, $d\mu_W = \sin\theta d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\mathbf{r}_i$. Every operator is an explicit differential operator, Hermitian by integration by parts.

Remark: the anomalous commutator — not needed

The traditional route to (4.41) (Klein 1929; Van Vleck 1951) derives the *anomalous* algebra of the body-fixed angular momentum,

$$[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma \quad (\text{note the } - \text{ sign}),$$

and uses it to control the operator ordering of the rotational kernel.

Nothing here used it

- $\hat{J}_\alpha$ entered only as the *shorthand* (4.24) for explicit Euler-angle derivatives.
- Hermiticity came from integration by parts + the divergence identity (4.26).
- The pseudopotential came from the Q -derivatives of the volume element.

The anomalous algebra is a correct property of the operators (4.24) — but it is a **consequence** of the coordinate representation, **not an input** to the quantization.

Born–Huang expansion and the BO truncation

At fixed Q , the clamped-nuclei electronic problem $\hat{H}_{\text{el}}(Q)|\phi_n(Q)\rangle = E_n^{\text{el}}(Q)|\phi_n(Q)\rangle$ (4.42–4.43) provides a complete parametric basis. The **exact** molecular wavefunction expands as

$$|\Psi\rangle = \sum_n |\chi_n\rangle \otimes |\phi_n(Q)\rangle \quad (4.45)$$

— not an ansatz: it follows from completeness at each Q . Projecting the Schrödinger equation gives coupled channels (4.46) with non-adiabatic couplings $\hat{\Lambda}_{nm}$ (4.47).

Born–Oppenheimer = diagonal truncation

Set $\hat{\Lambda}_{nm} \rightarrow 0$: single-surface nuclear dynamics (4.48) on the potential energy surface

$$V_n^{\text{BO}}(Q) = E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q). \quad (4.49)$$

A controlled approximation *of an exact operator* — breaking down at conical intersections.

Summary

The journey, in four stages

1. **Classical**: lab H (1.1) $\rightarrow$ one mixed mass-centered point transformation $\rightarrow H$ (2.12): COM exactly split for any $\mathbf{P}_{\text{cm}}$, mass polarization included — no boost, no rest frame.
2. **Body-Fixed**: Routhian (electrons stay canonical) $\rightarrow$ key identity $\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} \rightarrow$ classical H_{int} (3.22) with Watson $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}$.
3. **Quantize**, coordinate-natively: metric (4.15) $\rightarrow$ measure (4.21) $\rightarrow$ inverse (4.22) $\rightarrow$ Podolsky blocks $\rightarrow \gamma^{1/4}$ rescaling $\rightarrow$ Watson $\hat{U} = -\frac{\hbar^2}{8} \sum_{\alpha} \mu_{\alpha\alpha}$ (4.38). Hermiticity by integration by parts + divergence identity (4.26) — **no angular-momentum algebra**.
4. **Quantum $\hat{H}$** : the exact non-relativistic $\hat{H}_{\text{int}}$ (4.41).

What we did — and did not — assume

$\hat{H}_{\text{int}}$ (4.41) is **exact**. The only inputs were:

- non-relativistic dynamics;
- the Eckart frame (a convention);
- $3N_{\text{nuc}} - 6$ normal coordinates (an exact parametrization).

No Born–Oppenheimer, no clamped nuclei, no neglect of mass polarization, no harmonic assumption — and **no anomalous commutator as input**.

Outlook

(4.41) feeds the Born–Huang expansion (4.45), whose diagonal truncation recovers the familiar single-surface picture (4.48)–(4.49) — a controlled approximation *of an exact operator*.

**The full molecular Hamiltonian:
exact, quantized coordinate-natively,
and ready to approximate.**